MILITARY SPECIFICATION

GAGE, DEPTH, DIAL INDICATING: T4191A

1. SCOPE

1.1 This specification covers requirements, quality assurance provisons, and the preparation for delivery criteria for one type of gage, depth, dial indicating for general use.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for inspection by Attributes
MS 35276-228 - Screw

DRAWINGS

ORDNANCE CORPS

9210865	Gage, Depth
9210866	Plate, Support
9210867	Nut Locking
9210868	Support, Adjustable
9210869	Standoff
9210870	Point, Contact
9210979	Leg
9210980	Base
9216119	Carrying Case Assembly
9216128	Packing and Marking for Gage, Depth, Dial lG: XT4191A
9268153	Gage Depth Dial Indicating: T4191-A.
ET9268153	- Equipment Tabulation

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The gage shall comply with all requirements specified on the drawing (dwg) 9268153, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 Functioning. The dial gage readings shall be reproduceable when tested as specified in 4.4.1.

3.4 Workmanship,

3.4.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3,4.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing,

3.4.3 Burrs. Parts shall be free from burrs.

3.4.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material;

3.4.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. Not applicable.

4.3 Quality conformance inspection.

4.3.1 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose

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of this specification. The manner in which, each. inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. Critical characteristics shall require 102 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table I
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table II
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 4.3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Plate support (see dwg 9210866) covering a detail of dwg 9268153.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Location of tapped holes	Gage
102.	Pitch diameter of large tapped hole	Gage
103.	Minor diameter of large tapped hole	Gage
104.	Pitch diameter of small tapped holes	Gage
105.	Minor diameter of small tapped holes	Gage
106.	Thickness, min	Gage
107.	Material (see 3.1)	Visual
108.	Objective evidence (see 6.2) for protective finish in accordance with dwg 9210866	Visual
Minor:		
201.	Marking	Visual
202.	Workmanship (see 3.4)	Visual/Manual

4.3.2.1.2 Nut, locking (see dwg 9210867) covering a detail of dwg 9265153.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	pitch diameter of thread.	Gage
102.	Minor diameter of thread	Gage
103.	Material (see 3.1).	Visual
104.	Objective evidence (see 6.2) for protective finish in accordance with dwg 9210867. .,	Visual
Minor:		
201.	Workmanship (see 3.4)	Visual/Manual

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4.3.2.1.3 Support, adjustable (see dwg 9210868) covering a detail of dwg 9268153.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Diameter of small hole	Gage
102.	Location of small hole	Gage
103.	Depth of tapped hole (.74)	Gage
104.	Pitch diameter of tapped hole	Gage
105.	Minor diameter of tapped hole	Gage
	Pitch diameter of outside thread.	Gage
107.	Major diameter of outside thread.	Gage
108.	Length of outside thread	Gage
109.	Length to horizontal bar (2.26)	Gage
110.	Thickness of horizontal bar (.51)	Gage
111.	Length of collar (.385)	Gage
112.	Proper welding	Visual
113.	Material (see 3.1) .,	Visual
114.	Hardness	Gage
115.	Objective evidence (see 6.2) for protective finish in accordance with dwg 9210868.	Visual
Minor:		
201.	Marking	Visual
202.	Workmanship (see 3.4)	Visual/Manual

4.3.2.1.4 Stand off (see dwg 9210869) covering a detail of dwg 9268153.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Total length	Gage
102.	Pitch diameter of thread.	Gage
103.	Major diameter of thread	Gage
104.	Material (see 3.1).	Visual
Minor:		
201.	Workmanship (see 3.4)	Visual/Manual

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Table II
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 4.3.2) Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Plate cupport (see dwg 9210866) covering a detail of dwg 9268153

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Maj or:		
101.	Location of tapped holes	Gage
1020	Pitch diameter of large tapped holes	Gage
103.	Minor diameter of large tapped hole . . .	Gage
104.	Pitch diameter of small tapped holes. .	Gage
105.	Minor diameter of small tapped holes.. .	Gage
106.	Thickness, min.	Gage
107.	Material (see 3.1).	Visual
108.	Objective evidence (see 6.2) for protective finish in accordance with dwg 9210866	Visual
Minor:		
201.	Marking	Visual
202.	Workmanship (see 3.4)	Visual/Manual

4,3.2.1,2 Nut, locking (see dwg 9210867 covering a detail of dwg 9268153.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Pitch diameter of thread	Gage
102.	Minor diameter of thread	Gage
103.	Material (see 3.1)	Visual
104.	Objective evidence (see 6.2) for protective finish in accordance with dwg 9210867	Visual
Minor:		
201.	Workmanship (see 3.4)	Visual/Manual

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4.3.2.1.3 Support, adjustable (see dwg 9210868) covering a detail of dwg 9268153.

Categories		Characteristics	Method of Inspection
Critical:		None defined.	
Major:			
	101.	Diameter of small hole	Gage
	102.	Location of small hole	Gage
	103.	Depth of tapped hole (.74)	Gage
	104.	Pitch diameter of tapped hole	Gage
	105.	Minor diameter of tapped hole.	Gage
	106.	Pitch diameter of outside thread.	Gage
	107,	Major diameter of outside thread	Gage
	108.	Length of outside thread	Gage
	109.	Length to horizontal bar (2.26)	Gage
	110.	Thickness of horizontal bar (.51)	Gage
	111.	Length of collar (.385)	Gage
	112,	Proper welding	Visual
	113.	Material (see 3.1)	Visual
	114.	Hardness,	Gage
	115.	Objective evidence (see 6.2) for protective finish in accordance with dwg 9210868,	Visual
Minor:			
	201.	Marking	Visual
	202.	Workmanship (see 3.4)	Visual/Manual

4.3,2.1.4 Stand off (see dwg 9210869) covering a detail of dwg 9268153.

Categories		Characteristics	Method of Inspection
Critical:		None defined.	
Major:			
	101.	Total length.	Gage
	102.	Pitch diameter of thread	Gage
	103.	Major diameter of thread	Gage
	104.	Material (see 3.1)	Visual
Minor:			
	201.	Workmanship (see 3.4).	Visual/Manual

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4.3.2.1.5 Point, contact (see dwg 9210870) covering a detail of dwg 9268153.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Pitch diameter of thread	Gage
102.	Major diameter of thread, ...	.Gage
103.	Length of thread (.188)	.Gage
104.	Total length (2. 198)	.Gage
105	Outside diameter (.156)	.Gage
	Material(see 3.1)	Visual
Minor:		
201.	Workmanship (see 3.4)	Visual/Manual

4.3.2,1.6 Leg (see dwg 9210979) covering a detail of dwg 9268153.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Chamfer present	Visual
102.	Width (.500)	Gage
103.	Pitch diameter of threaded hole.	Gage
104.	Minor diameter of threaded hole	.Gage
105.	Depth of threaded hole (.18)	Gage
106.	Location of threaded hole	Gage
107.	Length from point to top (.813).	.Gage
108.	Length (.77)	.Gage
109.	Material (see 3.1)..	Visual
Minor:		
201.	Workmanship (see 3. 4)	..Visual/Manual

4.3.2.1.7 Base (see dwg 9210980) covering a detail of dwg 9268153.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Width of slots (.500)	.Gage
102.	Depth of slots (.26)	.Gage

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Major:
103. Location of small holes . . . . . .Gage
104. Diameter of small holes (.154) . . . . .Gage
105. Pitch diameter of threaded hole . . . . . Gage
106. Minor diameter of threaded hole.. . .Gage
107, Location of threaded hole . . . . . Gage
108. Material (see 3.1) . . . . . Visual
109. Objective evidence (see 6.2) for protective finish in accordance with dwg 9210980. . . . . .Visual

Minor:
201. Workmanship (see 3,4). . . . . Visual/Manual

4.3.2.1.8 Gage depth, dial indicating: T4191-A (see dwg 9268153).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Ability of contact point and dial pointer to move freely (without binding) when actuated by lifting lever.	..Visual/Manual
102.	No components missing.	.Visual
103.	Components properly assembled	.Visual
104.	Objective evidence (see 6,2) for gage in accordance with dwg 9210865	.Visual
105.	Objective evidence (see 6.2) for screw in accordance with MS 35276-228.	.Visual
Minor:		
201.	Workmanship (see 3.4)	.Visual/Manual

4,3.2,1,9 Carrying case assembly (see dwg 9216119) covering a detail of dwg 92161250

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Cover located symmetrically with base.	.Gage
102.	All parts present.	.Visual
103,	Component parts mate correctly	.Visual/Manual
104.	Parts properly secured	.Visual/Manual

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105. Objective evidence (see 6.2) for protective finish in accordance with dwg 9216119. . . . . .Visual
106. No paint missing. . . . . Visual
107. Location of handle and latches. . . . . Visual

Minor:
201. Workmanship (see 3.4). . . . . Visual/Manual

4.3.2.1.10 Packing and marking for gage, depth, dial 1G: XT4191A (see dwg 9216128)

Categories	Characteriastics	Method of Inspection
Critical:	None defined.	
Major:		
101,	Material (see 3.1)	Visual
102.	Parts correctly packed.	Visual/Manual
103.	Contents tight in carton	.Manual
	Carton properly sealed	.Visual
Minor:		
201.	Marking	.Visual
202.	Workmanship (see 3 . 4)	.Visual

4.3.3 Testing. This test shall be performed.

	Test	Test Classification
a.	Functioning	Major

This test is not destructive and samples so tested may be returned to the lot.

4,3.3.1 Functioning. Each gage shall be tested in accordance with 4.4.l. If any item fails to comply with 3.3, the item shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4,1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions> and calibration instructions specified on the Equipment Lists (EL). assemblies are listed on the Equipment Tabulation, ET, When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "OD" is listed in the "Part or "Identifying Number" column on the EL (see 6.1.2).

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4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accurancy tolerances of the test equipment used.

4.4.1 Functionting (see 3.3). Set the T4191 dial indicating gage on a clean reference surface plate which is flat within 10 percent of the indicator graduation value. Adjust the indicator position so at least one complete revolution is made at the "at rest" position to assure positive contact. Use a master (Jo block) whose thickness is known and is within the full range of the indicator. Set the indicator dial to zero and check that the indicator measures correctly the thickness shown on the master. Remove the (Jo block) and check zero reading. Replace the (Jo block) and check the repeatability.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The gage shall be packaged in accordance with dwg 9216128 using the specified moisture barrier.

5.1.2 Level C. The gage shall be packaged in accordance with dwg 9216128 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The gage shall be packed in accordance with dwg 9216128.

5.3 Marking. Marking shall be as specified in accordance with dwg 92161280

6. NOTES

6.1 Ordering data. Procurement documents should specify:

6.1.1 Procurement requirements.

a. Title number, and date of this specification.

b. Selection of-applicable level of packaging and packing required (see 5.1 and 5.2)

6.1.2 Contact data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1667 or specification when specified on a DD Form 1423 incorporated in the contract.

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6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-NR4.

Custodian:
Army - MU

Preparing activity:
Army - MU

Project No. 1190-A059

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-G-50962(MU) Gage, Depth, Dial Indicating: T4191A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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DEPARTMENT OF THE ARMY

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-NR
Dover, New Jersey 07801

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NOTICE OF INACTIVATION FOR NEW DESIGN

INCH-POUND

MIL-G-50962 (AR)
NOTICE 3
21 June 1995

MILITARY SPECIFICATION

GAGE, DEPTH, DIAL INDICATING: T4191A

This notice should be filed in front of MIL-G-50962 (AR), dated 15 October 1971

MIL-G-50962 (AR) dated, 15 October 1971 with Validation Notice 2, dated 20 August 1991 is inactive for new design and is no longer used, except for replacement purposes.

Preparing Activity
Army-AR

AMSC N/A

FSC 1190

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